

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (IT)

May 2022

IT379: Computer Vision

Time: 10:00 am To 01:00 pm

Maximum Marks: 70

Date: 14.05.2022, Saturday

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of a scientific calculator is allowed.

SECTION – I**Q - 1 Choose any one from multiple options.****[05]**

(a) Name the procedure in which individual pixel values of the digital image get altered. **[01]**

1. Neighborhood Operations
2. Image Registration
3. Geometric Spatial Transformation
4. Single Pixel Operation

(b) Using the feature of the reciprocal relationship of filter in the spatial domain and corresponding filter in the frequency domain, which of the following fact is true? **[01]**

1. The narrower the frequency domain filter results in increased blurring
2. The wider the frequency domain filter results in increased blurring
3. The narrower the frequency domain filter results in decreased blurring
4. None of the mentioned

(c) What is the relation between the blurring effect and with change in filter size? **[01]**

1. Blurring increases with a decrease in the size of filter size
2. Blurring decreases with the decrease of the size of filter size
3. Blurring decreases with the increase of the size of filter size
4. Blurring increases with the increase of the size of filter size

(d) In general, which of the following assures no ringing in the output? **[01]**

1. Gaussian Low pass Filter
2. Ideal Low pass Filter
3. Butterworth Low pass Filter
4. All of the mentioned

(e) The general form of log transformations is _____. **[01]**

1. $s = c \cdot \log(1 + r)$
2. $s = c + \log(1 + r)$
3. $s = c \cdot \log(1 - r)$
4. $s = c - \log(1 - r)$

- Q - 2 Write the answer to the following questions.** [14]
 (a) Give various definitions of Computer Vision. [03]
 (b) Define image sampling and image quantization. [02]

OR

- (b) Why there are so many Color models like RGB, CMY, HSV, etc.? [02]
 (c) Consider a 3-bit image of size 64*64 pixels has intensity distribution shown in the table. Show each step of calculations with correct histograms of the original and equalized images. Also, evaluate the equalized image after the transformation steps. [07]

Gray Levels	0	1	2	3	4	5	6	7
No. of Pixels	190	250	210	160	80	60	30	20

- (d) What is geometric transformation? [02]

OR

- (d) Specify the properties of the 2D Fourier transform. [02]

- Q - 3 Write the answer to the following questions.** [16]

- (a) (1) Obtain the image negative of the following 8 bits per pixel image. [04]
 (2) Perform Gray Level Slicing using as per below-given approach on below-given image and generate an output image.
 Approach: In Input Image, all intensity values from 0 to 127 turn black, and other turns white.

124	205	169	156	151
139	127	157	117	125
252	147	236	138	145
227	182	178	197	242
201	106	119	235	240

- (b) Find all the bit planes of the following 3-bit image: [02]

4	1	3	7
1	6	2	5
2	3	6	4
1	2	5	3

- (c) Explain what camera calibration is. What are the steps involved in the procedure? [05]
- OR**
- (c) Discuss the significance of depth estimation with various techniques in computer vision. [05]
- (d) Consider the following image given below. What will be the new value of pixel at (3, 3) if below stated smoothing operations is done using 3*3 neighborhood. [05]
- (a) Mean Filter
- (b) Weighted average Filter
- (c) Median Filter
- (d) Min Filter
- (e) Max Filter

4	65	72	7	1
13	55	25	41	64
85	10	11	81	33
1	45	4	7	12
15	21	20	90	76

OR

- (d) Let's take an image of the dimensions of 3 rows and 3 columns and zoom it twice using any method. [05]

7	8	5
6	10	9
5	7	4

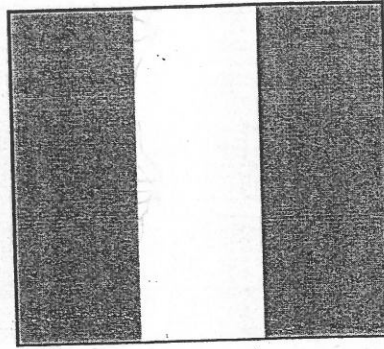
SECTION – II

- Q - 4** Write the answer to the following questions. [05]
- (a) First derivatives in image segmentation produce _____. [01]
1. Thick edge
 2. Thin edge
 3. Fine edge
 4. None of the above
- (b) Define Erosion. [01]
- (c) What do understand by boundary extraction? [02]
- (d) List down Feature Detection And Matching techniques [01]

- Q - 5 Write the answer to the following questions.** [14]
 (a) Explain the three fundamental steps performed in edge detection. List down the techniques to achieve the objectives for edge detection? State applications of edge detection. [07]

OR

- (a) How an edge of the image is detected? Discuss any edge detection algorithm for detecting an edge of the following image. [07]



- (b) Explain Region Splitting and Merging. [05]
 OR
 (b) Explain about basic thresholding process used in image segmentation. [05]
 (c) Differentiate SIFT and SURF. [02]
 OR
 (c) What is HOG (Histogram of Oriented Gradients)? [02]

- Q - 6 Write the answer to the following questions.** [16]
 (a) How is supervised learning different from unsupervised learning. Explain with an example. [04]
 (b) What is an active contour model? How contours are represented using this model? [05]
 OR
 (b) Short Note on 3D Reconstruction in computer vision. [05]
 (c) Briefly explain Hough transform technique to detect circle shape. Assume an equation for circle by $(x-a)^2 + (y-b)^2 = R^2$; where (a, b) is the centre of the circle and R is known radius. [07]